

EVORA AP, Portugal

WMO: 085580

Lat: 38.5367N

Lon: 7.8881W

Elev: 246

StdP: 98.41

Time Zone: 0.00 (EUW)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.1	2.4	-5.4	2.5	7.6	-3.0	3.0	7.7	10.8	10.8	9.6	11.0	1.9	0	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.6	37.6	19.7	35.8	19.4	34.0	18.9	21.2	33.1	20.5	31.9	19.9	30.6	2.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.3	13.6	21.7	17.6	13.0	21.5	16.9	12.4	21.3	62.3	33.2	60.0	32.0	57.8	30.6	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.1	7.9	7.0	-1.4	40.6	1.4	1.9	-2.4	42.0	-3.2	43.1	-4.0	44.2	-5.0	45.6	
			DB												
			WB	-2.4	22.2	1.4	0.8	-3.5	22.8	-4.3	23.2	-5.1	23.7	-6.2	24.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	9.4	10.3	12.6	14.3	17.7	22.0	24.2	24.7	21.9	18.0	12.6	10.0
	DBStd	6.16	2.44	2.48	2.73	3.03	3.54	3.55	2.91	2.83	2.78	3.00	2.79	2.44
	HDD10.0	112	41	24	7	2	0	0	0	0	0	0	8	30
	HDD18.3	1343	278	223	177	127	54	8	1	0	3	42	173	257
	CDD10.0	2488	21	34	89	130	241	359	439	454	357	248	85	31
	CDD18.3	677	0	0	1	4	36	117	181	196	110	31	1	0
	CDH23.3	8268	0	0	13	82	504	1511	2278	2436	1174	268	2	0
	CDH26.7	4155	0	0	0	11	183	764	1253	1348	517	78	0	0
Wind	WSAvg	3.0	2.8	3.0	3.2	3.2	3.0	3.3	3.5	3.2	2.7	2.7	2.8	2.7
Precipitation	PrecAvg	638	88	80	54	57	48	26	7	5	28	72	82	89
	PrecMax	1040	342	208	209	175	121	105	73	73	77	217	245	248
	PrecMin	367	1	4	0	4	0	0	0	0	1	2	4	4
	PrecStd	178	79	63	49	39	36	28	15	13	22	60	67	74
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	19.9	25.1	28.1	33.8	38.4	39.8	39.6	36.5	32.1	23.0	18.4
		MCWB	12.7	13.6	14.6	16.8	18.2	19.7	19.9	20.4	19.2	17.0	15.8	13.7
	2%	DB	15.9	18.0	23.0	25.7	31.1	35.9	37.6	37.6	33.9	28.7	20.6	16.4
		MCWB	11.9	12.4	13.9	16.2	17.5	19.3	19.4	20.0	18.4	16.6	14.6	13.9
	5%	DB	14.6	16.5	21.0	23.6	28.6	33.7	35.5	35.7	32.0	26.2	18.7	15.4
		MCWB	11.5	12.1	13.6	15.3	16.9	18.8	19.1	19.7	18.0	16.5	14.5	13.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	13.6	15.0	18.7	21.2	26.1	31.3	33.4	33.6	29.8	24.0	17.3	14.3
		MCWB	11.3	11.5	13.0	14.4	16.4	18.4	18.9	19.0	17.7	16.0	13.9	12.3
	0.4%	WB	14.5	15.1	16.7	18.7	20.2	21.6	21.7	22.1	21.0	20.2	18.3	16.1
		MCDB	16.1	17.8	22.6	26.2	29.4	33.5	33.3	35.0	32.7	24.3	19.8	16.6
	2%	WB	13.5	13.9	15.4	17.1	18.9	20.6	20.9	21.3	20.0	19.1	16.7	14.8
		MCDB	14.7	16.4	20.3	23.5	27.6	32.4	32.7	33.7	29.3	23.4	18.5	15.9
Mean Daily Temperature Range	5% DB	WB	12.7	13.0	14.4	15.8	17.9	19.8	20.2	20.6	19.3	18.2	15.6	13.8
		MCDB	13.8	15.5	19.1	22.0	26.0	31.1	32.0	32.4	27.7	22.5	17.5	14.9
	5% WB	WB	11.7	12.2	13.5	14.9	17.0	18.9	19.5	19.8	18.6	17.4	14.8	12.8
		MCDB	13.1	14.6	17.7	20.2	24.5	29.1	31.1	30.7	26.4	21.6	16.6	14.1
	Clear-Sky Solar Irradiance	taub	0.312	0.328	0.359	0.368	0.376	0.379	0.359	0.376	0.372	0.350	0.325	0.313
		taud	2.504	2.458	2.379	2.361	2.367	2.373	2.404	2.367	2.399	2.490	2.516	2.511
Ebn at Noon		869	899	900	910	904	897	913	890	874	863	846	840	
Edh at Noon		78	93	111	120	122	121	116	118	108	89	77	73	
All-Sky Solar Radiation	RadAvg	2.32	3.27	4.38	5.51	6.65	7.54	7.82	6.93	5.34	3.67	2.57	2.06	
	RadStd	0.30	0.40	0.49	0.52	0.59	0.42	0.24	0.24	0.34	0.32	0.32	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page